Multimedia Tools and Applications (2023) 82:10703–10722
https://doi.org/10.1007/s11042-022-13709-y

Text line segmentation in indian ancient handwritten documents using faster R-CNN

Amar Jindal[1] · **Rajib Ghosh**[1]

Received: 7 August 2021 / Revised: 14 April 2022 / Accepted: 24 August 2022 /
Published online: 12 September 2022
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Abstract
Textline segmentation in ancient handwritten documents is still considered as a challenging task in document analysis and recognition field even though various rule-based methods exist. These methods succeed under constraint such as a roughly uniform background. They do not contribute well in case of variable inter-line spacing and overlapping characters. This article proposes faster region-convolution neural network (R-CNN) based robust method to segment the textlines in the ancient handwritten document in Devanagari script for the first time in literature. The feature matrix has been generated by residual network and proposals have been predicted through the region proposal network (RPN). A pooling layer has been used to extract regions of interest, known as region of interest pooling layer, to locate the textlines. The performance of the proposed textline segmentation system has been evaluated on self generated dataset of ancient handwritten documents in Devanagari script and it has achieved the f-measure of **99.98%**. Experimental results demonstrate that the proposed system outperforms the existing state-of-the-art methods of textline segmentation.

Keywords Text line segmentation · Ancient documents · Devanagari script · Faster R-CNN

1 Introduction

Any handwritten text recognition task can be performed in two different modes—online and offline. In the online mode of recognition, the text is written with the help of a pen like stylus on a tablet and the handwritten text is instantly recognized by the system at the very moment of writing it on the tab. In this recognition mode, movement of pen-tip and time series of coordinates are collected during text collection [5, 11]. In offline mode, the text is written on a piece of paper and the image of the document is fed to the computer through optical scanning. The text present in the scanned image is recognized by the system later [4, 21, 22]. In the recent decades, intensive works are going on offline handwritten text recognition (HTR) in ancient historical documents [7, 14, 18]. In the southern part of the Asia

✉ Rajib Ghosh
rajib.ghosh@nitp.ac.in

[1] Department of Computer Science and Engineering, National Institute of Technology Patna, Patna, 800005, India

continent, countries like India, Pakistan, Nepal, Bangladesh, Sri Lanka, etc. are filled with ancient historical documents in several scripts. These documents are multilingual in nature using the same script. Ancient documents are rich with knowledge of science, medication, religious, political, and other different historical kinds of literatures. These documents are having numerous preservation challenges and digitizing them is one of the solutions to it. An optical character recognition (OCR) model includes segmentation of text/non-text portions, textlines, words, and characters in multi-sentenced document. After the segmentation task, generally the character recognition and word recognition task are performed [7, 14, 18]. However, recognition of the handwritten text in ancient historical documents is not an easy task and faces several challenges as mentioned below.

1. The colour of ink starts vanishing with time. As a result, the text gets faded in the documents. One of such document images is presented in Fig. 1.
2. The ancient documents consist of dark spots due to the absorption of ink of other documents as well as humidity. Two such documents are shown in Figs. 2 and 3.
3. As the years pass, the quality of the papers on which text is written gets degraded and leads to a page torn. One of such situations is illustrated in Fig. 4.
4. The text skewness, varying layouts, variable interline spacing, broken characters, overlapping of textlines as well as characters create other recognition challenges like under-segmentation and over-segmentation of textlines [22]. Figures 1, 2, and 4 show few ancient historical handwritten documents with text skewness, varying layout, and interline overlapping respectively.

A textline can be defined as ruling to the text which shows the position and presence of the text within a page. It is also used for layout analysis of the document and basis enhancement of the readability. Textline segmentation (TLS) is the very first step to start building any OCR for multi-sentenced documents. So, several investigations have been reported [3, 10, 13, 15, 17, 19, 21, 22, 24, 29] to segment the textlines in ancient handwritten documents, but most of these investigations have been carried out in Latin, Chinese, Arabic, and Japanese scripts. These existing studies have mostly relied upon various conventional techniques like projection profile [21, 22], component grouping [13, 15, 24], and hough transform [17] based methods to segment the textlines in various ancient documents. But these conventional techniques are not capable to overcome all the challenges like stains of ink and humidity on text, faded text, and inter-connection of two consecutive lines present in the ancient historical documents because these techniques follow a fixed mathematical approach and require prior knowledge of the layout structure of the documents. The conventional techniques consume more time to extract features at all the points of the document image. Few of the conventional techniques such as projection profile based methods fail to segment the textlines correctly if the interline overlapping and skewness of the text are present in the document because these methods cannot find any row containing white pixels

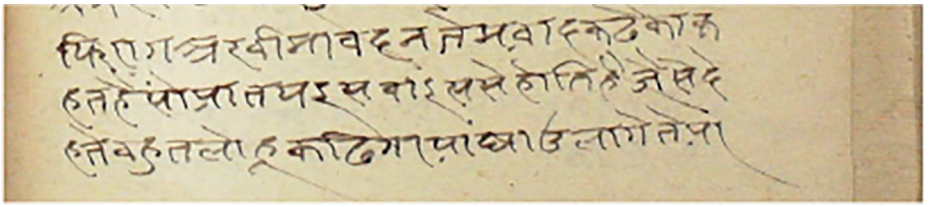


Fig. 1 Faded text and skewed textlines in Ved vedant manuscript document image

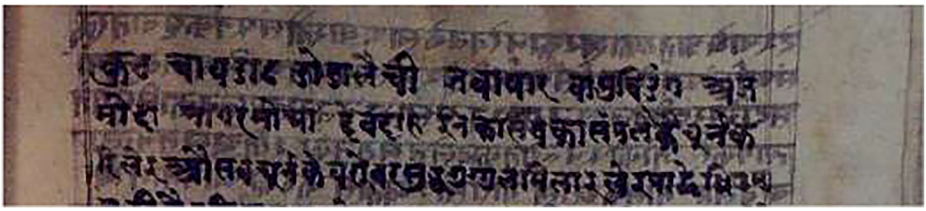


Fig. 2 Absorbed ink stains and varying layout in religious manuscript document image

only. The decline of TLS accuracy can be seen in grouping based methods as well because the faded characters fail to contribute to the calculation of connected components.

This article proposes a faster region-convolutional neural network (R-CNN) [26] model based method for TLS in ancient historical handwritten Devanagari document images for the first time in the literature. The present work has used the residual network Resnet50 as a backbone architecture in faster R-CNN for the first time in this research domain (TLS in ancient handwritten Devanagari document images). Faster R-CNN is a deep learning architecture used for object detection using a region proposal algorithm. Faster R-CNN provides very high object detection accuracy in comparison to other convolution neural network (CNN) based architectures due to the generation of various region proposals using the region proposal algorithm. It drastically reduces the training time and testing time for the detection network. By exploiting the capability of the faster R-CNN detection network, the textlines of the ancient handwritten historical documents can be detected easily because textline can also be seen as a bounding box [20] or a set of connected text pixels [31]. Moreover, if the problems like faded text, interline overlapping, and skewness of the text are present in the documents, then the existing methods fail to solve these problems, whereas the proposed faster R-CNN based method resolves these problems in most of the samples except few as it relies on the generation of various region proposals using the region proposal algorithm.

The contributions of the proposed work are given below.

1. Proposal of modified faster R-CNN model, a deep learning method, for segmenting the textlines in ancient historical handwritten documents in Devanagari script.
2. The Residual network (Resnet50) backbone architecture has been used in faster R-CNN to segment the textlines in the ancient historical handwritten documents in Devanagari script.
3. The inclusion of 5 different scaling factors of the anchor boxes by modifying the existing scaling factors of the conventional faster R-CNN.
4. Evaluation of state-of-the-art methods in foreign scripts on Devanagari ancient historical documents used in the present work.

The remaining portions of the paper is organized as follows: the different ad-hoc and machine learning based investigations of textline segmentations are discussed in Section 2.

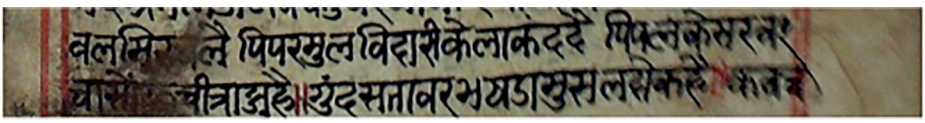


Fig. 3 Dark spot and varying layout present in religious manuscript document image

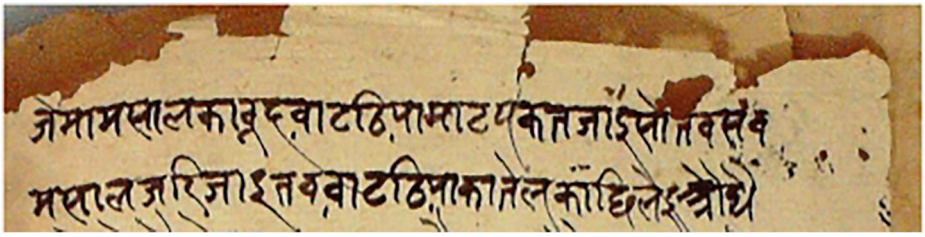


Fig. 4 Page torn and interline overlapping in ancient Ayurveda document image

The dataset description is provided in Section 3. Section 4 presents the implementation details of faster R-CNN. Evaluation metrics and comparison with different approaches are stated in Section 5. Finally, Section 6 infers our proposed work.

2 Related work

As already mentioned a little number of studies are available for handwritten text recognition in the ancient historical document in Devanagari script. Most of the studies have focused on various non-Indian scripts. These existing studies in both Indian and non-Indian scripts are discussed below briefly under two different categories, Ad hoc and machine learning based methods.

2.1 Ad hoc methods

Narang et al. [21] have developed a complete ancient Devanagari handwritten text recognition system on self generated data. For textline segmentation, the piece-wise projection profile method has been used where an image has been divided into fixed strips and then the horizontal projection profile has been applied in each strip. The problem of under-segmentation and over-segmentation has been resolved by computing the average height of the textline for each piece-wise strip and then applying the projection profile method in another study [22] on ancient historical handwritten document in Devanagari script. The projection profile based methods are not able to provide a very high TLS accuracy when a large amount of noise is present, variable inter-line spacing exists, and touching characters are present in almost every textline of the document image. Similarly, the seam curving between the two adjacent lines has been used to segment the textlines in ancient historical documents. To find the medial seam, Saabni et al. [29] have divided the image into a fixed number of strips and then the Sobel operator has been applied to observe the horizontal seam.

The grouping methodology involves the alignment of text units by clustering them in a bottom-up approach. Black pixels of primitive components i.e. text characters are grouped to form a unit. These connected components are further joined to find out textline alignment. The challenges of this bottom-up grouping strategy like identifying factual alignment of textline, overlapping textline, and choosing nearest neighbour have been resolved by iterative perceptual grouping in [15]. In another study [13], the average height and width of each character in the textline are considered to be equal. The initials of the textline are identified by the average height threshold of the connected component. But, the major drawback of this method is it fails if the average height and width of the character are not equal. Garz

et al. [3] have used the grouping methodology without performing binarization on the document image samples from the Saint-Gall dataset. The clustering interest points have been extracted using gaussian difference at locations of local minima and local maxima which represent different portions of the character.

Rabaev et al. [24] have extracted the potential characters to identify a textline using connected components. The sweep line algorithm has been applied to find out the position of each potential character element to identify a textline. In another study [10], on one of the Indic scripts, both text and non-text portions were present in the same ancient historical documents. The study applied the grouping clustering technique to perform the segmentation of text and non-text. Two clusters were formed because the space difference between the text and non-text portions is larger as compared to each of them. However, the grouping based methodology is not able to calculate the correct component size due to the pixels of faded character.

Messaoud et al. [19] proposed a combination of three different methods as a multilevel TLS framework. At the first level, the low threshold projection profile method was applied for samples not containing the interline overlapping of textlines and horizontal textlines. For the small amount of touching components, the grouping methodology was used as a second level TLS. In the third level, the huge overlapping between the interline words was resolved by calculating the nearest neighbour of the words. Louloudis et al. [17] have initially calculated the connected components and average height of the character. The subsets of connected components were created based on the fixed threshold of the average height and average width of the character and components. The Hough transform mapping was applied to each subset. In the final step of the post-processing, the subsets were merged into identified textlines. To identify the textlines from the large connected components, the skeleton operation was performed on connected components.

2.2 Machine learning based methods

Apart from Ad hoc methods, few studies are available using various machine learning based methods also. In a recent study [14] towards TLS in historical documents in Latin script, a fully convolutional network (FCN) based architecture has been proposed to detect the baselines. The horizontal projection profile method has been applied to the detected baseline to calculate the height of the character. Finally, the textline has been cropped by after adding some margin to the height of the character. However, FCN architecture based methods also learn the noises like ink stain, humidity spots, etc. present in the document due to ageing of the document. Moysset et al. [20] used the recurrent neural network (RNN) to segment the paragraph and textlines of historical documents. Only the count of textlines in each paragraph was provided in the dataset used by the study. The RNN took the document image vertically to detect horizontal lines of characters. The skewness in the textline is the layout challenge of the TLS as well as the HTR. Vo et al. [31] obtained the line map using the FCN model and the line adjacency graphs were used to classify the touching text components into textlines. However, this method cannot identify a textline properly if skewness is present as it does not generate any region of interest (ROI) using region proposal technique.

In the HTR system, each pixel of the document image belongs to either foreground or background. Grüning et al. [6] used this concept and applied the FCN based architecture to classify the pixel label. The clustering approach was then used to build baselines. In another study [1], convolutional recurrent neural network (CRNN) based architecture with visual geometry group (VGG)-16 [16] as a backbone network was proposed to segment the textlines in ancient historical documents in Latin script. The training follows incremental

approach with only 10% of manually annotated data. The feature matrix were generated by convolution operation followed by the long short-term memory (LSTM) to detect the baselines. Renton et al. [27] applied FCN with dilated convolution to calculate the *x*-height of the core text to identify the textline. The FCN resulted in a spatial feature heatmaps in low resolution of the input document image. The pixel-level labelling was applied to the original resolution input image. To reflect the features of the input image with original resolution in the spatial feature heatmaps, the dilating convolutions, unpooling, and deconvolution operations could be the solutions.

The survey shows that the existing studies are not capable to resolve all the layout, skewness, interline overlapping, and preservation challenges of ancient handwritten historical documents. But, the proposed faster R-CNN architecture based method can overcome all the above said challenges as it relies on generation of various ROIs using region proposal network (RPN).

3 Dataset description

In this work, a new dataset on ancient handwritten historical Devanagari documents has been generated as no dataset is publicly available in this problem area. The ancient historical document images are in very delicate condition. So, developing a dataset of such type of documents is not an easy task. The documents have been collected from various libraries, museums, and archaeological departments. The documents collected in this dataset are different manuscripts and scriptures in Devanagari script of varying fields from 15th to 18th century originated from central India. The Devanagari script has been widely used to write different Indo-Aryan languages like Hindi, Marathi, Sanskrit, Sindhi, Konkani, Bundelkhandi, Nepali, Bodo, and Maithili. The documents have been collected in image form using a digital camera. Ayurveda, Religious, Astrological, Vastu, Vedic literature, Tantra, Ved-vedant, and Political history are the fields of different manuscripts and scriptures collected to generate this dataset. The text skewness and interline overlapping are present in almost every document images of all the categories of the generated dataset, whereas the page torn, faded text, and dark spots due to ink or humidity are present mainly in the samples of Ayurveda, Ved-vedant, Tantra, and Religious manuscripts. Figures 5, 6, 7 and 8 show four different ancient handwritten document images in Devanagari script in the fields of Ayurveda, Ved-vedant, and tantra respectively. The document samples presented Figs. 6, 7 and 8 containing the dark spot, faded text, and torn page respectively.

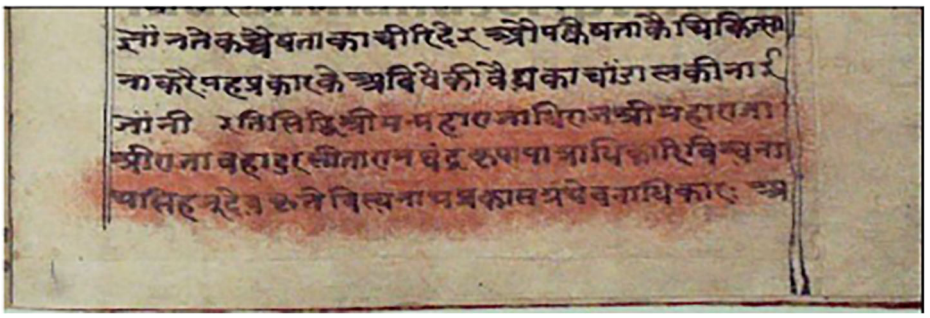


Fig. 5 A document image sample from Ayurveda field

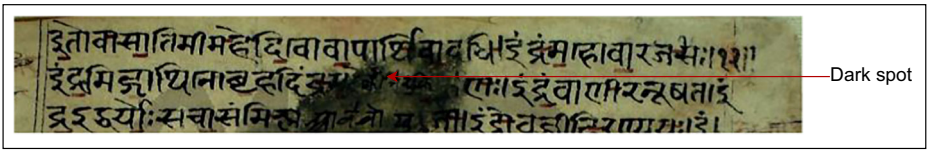


Fig. 6 A document image sample from Ved-vedant field containing dark spot

A total of 19889 ancient handwritten document image samples from the aforementioned fields have been collected. The dimension of each collected document is of size 800 x 800. The field wise document collection statistics is presented in Table 1. The collected documents have been further divided into training and testing sets with 7:3 ratio. Analysis has shown that varying numbers of textlines are present in the collected ancient handwritten document images. Figure 9 shows the distribution of number of document images present in the dataset against varying numbers of textlines present in all the collected documents.

4 Proposed method

The present investigation proposes an end-to-end method for TLS in ancient historical handwritten documents by proposing a modified form of the conventional faster R-CNN structure. Ren et al. [26] proposed the conventional faster R-CNN for object detection. The faster R-CNN can detect an object of varying sizes with various simple as well as complex backgrounds and in various adverse situations. This robust behaviour of faster R-CNN has been used to overcome the various challenges of TLS such as text skewness, dark spots, and torn pages present in the ancient document images in the present work.

The conventional faster R-CNN uses the VGG-16 as a backbone network to extract the spatial features, whereas the proposed modified form of faster R-CNN structure has used Resnet50 [9] as a backbone network as the present system has delivered the better performance (the results may be seen in Table 8) using this backbone network in comparison to using VGG-16 as a backbone network in faster R-CNN structure. Apart from this, the conventional faster R-CNN uses nine different anchor boxes whereas the proposed modified form of faster R-CNN has used fifteen different anchor boxes. The output of the Resnet50 is a feature matrix of size 25 x 25 x 2048. The proposed architecture of faster R-CNN produces various region proposals using a single RPN utilizing the feature matrix produce by Resnet50. An additional ROI pooling layer has been used to select some of the region proposals out of all generated by the RPN. The proposed modified architecture of the faster R-CNN is presented in Fig. 10.

The proposed architecture consists of the following steps:

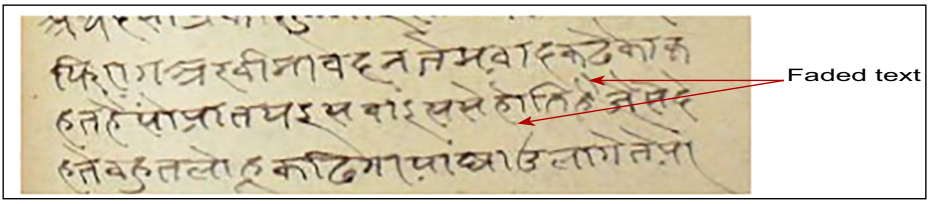


Fig. 7 A document image sample from Ved-vedant manuscript containing faded text

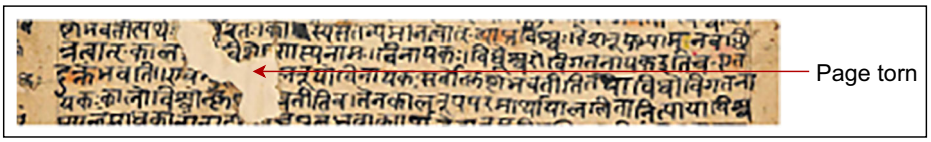


Fig. 8 A document image sample from Tantra field where the page is torn

1. Input the document image sample to the backbone network Resnet50.
2. Creation of several ROIs from the feature matrix obtained from Resnet50 using the RPN.
3. Pooling of ROIs obtained from step 2 using a separate ROI pooling layer.
4. Detection of textlines using two fully connected layers based on the trained model generated during training of the RPN.

4.1 Backbone network

The Resnet50 has been used as a backbone network in the proposed modified model of faster R-CNN. A 3D feature matrix has been obtained from the last layer of Resnet50 corresponding to the input document image. In the present investigation, Resnet50 consists of 49 convolution layers divided into four stages, whereas the conventional Resnet50 architecture contains 50 different layers. The fully connected layer of conventional Resnet50 has not been used in the present investigation. Each stage contains varying numbers of convolution blocks, where each block contains three convolution layers. The detailed statistics of the number of convolution blocks in each stage as well as the number of filters used in each layer are presented in Table 2.

In Resnet50, the feature matrix generated after a nth convolution layer is merged with the matrix generated from $(n{+}3)^{th}$ layer, where n varies from 1 to 46 with a step size of 3. This characteristic of Resnet50 provides residual behaviour to the backbone network. The Resnet50 provides faster convergence to the proposed modified model of faster R-CNN in comparison to the conventional faster R-CNN. It also shows the superior performance and lesser training time of the proposed modified form of the faster R-CNN in comparison to the conventional faster R-CNN model (Section 5 may be seen).

Table 1 Dataset statistics

Manuscript category	Total sample	Training sample	Testing sample
Ayurveda	3756	2630	1126
Religious	1984	1389	595
Astrological	1897	1328	569
Tantra, mantra, yantra	2542	1780	762
Vastu	1937	1366	571
Vedic literature	1913	1340	573
Ved vedant	2978	2085	893
Political history	2882	2018	864

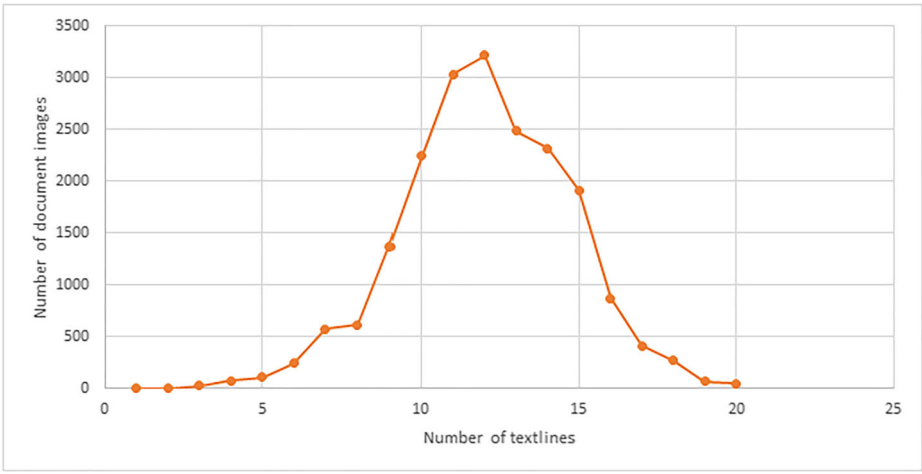


Fig. 9 Distribution of number of documents against varying numbers of textlines present in all the documents in the generated dataset

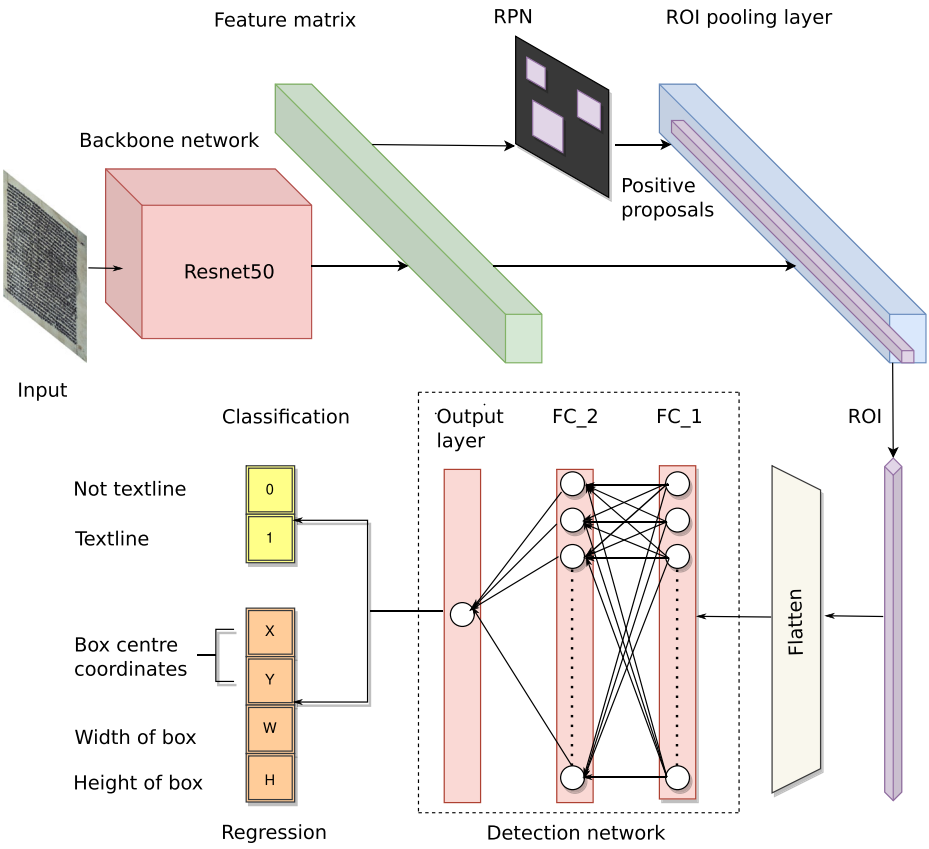


Fig. 10 Proposed architecture of faster R-CNN for textline segmentation

Table 2 Blocks and layers of Resnet50

Stage number	Number of convolution blocks	Kernal size	Number of filters
Conv_1	1	7 x 7	64
Stage_1		1 x 1	64
	3	3 x 3	64
		1 x 1	256
Stage_2		1 x 1	128
	4	3 x 3	128
		1 x 1	512
Stage_3		1 x 1	256
	6	3 x 3	256
		1 x 1	1024
Stage_4		1 x 1	512
	3	3 x 3	512
		1 x 1	2048

4.2 Region proposal network

The RPN uses a small network where a single window slides over the convolutional feature matrix obtained from the last layer of Resnet50 to generate the region proposals. Each sliding window has been mapped to 256-D Zeiler and Fergus network [33] to obtain spatial features for every location. The RPN predicts multiple region proposals for each location of the feature matrix simultaneously. Each region proposal is known as an anchor box. As many anchors are there in one sliding window, that many region proposals are generated in each location of the feature matrix. The centre point of the anchor box coincides with the centre point of each sliding window. Different aspect ratios and scales are associated with these anchors. In the present work, three different scales and five different aspect ratios, yielding a total of fifteen anchor boxes for each location have been used. On the other hand, the conventional faster R-CNN uses a total of nine different anchor boxes with different aspect ratios. The total number of anchor boxes for the convolutional feature matrix of size H x W x k, where (H,W) denotes the total number of rows and columns of the convolutional feature matrix respectively and k is the total number of anchors generated for each location of the feature matrix. The statistics of different anchor boxes used in the present investigation are presented in Table 3.

The aforementioned multi-scale anchors at every location of the output feature matrix has allowed us to avoid the extensive selective search [30] complexity to compute the proposals from the feature matrix. The positive labels have been assigned to the anchors satisfying the following condition:

Table 3 Statistics of different anchors

Scales	Size (height x width)				
1:1	32 x 32	64 x 64	128 x 128	256 x 256	512 x 512
1:2	32 x 64	64 x 128	128 x 256	256 x 512	512 x 1024
2:1	64 x 32	128 x 64	256 x 128	512 x 256	1024 x 512

Condition: If the intersection-over-union (IoU) [28] overlap value between the anchor box and ground truth (GT) box is greater than 0.88, then assign positive label to the anchor box.

Bayesian optimization technique [23] has been used to find the optimal IoU value 0.88. All the positive labelled anchor boxes generated by RPN have been fed as input to the ROI pooling layer.

4.3 Region of interest pooling

In the present work, the RPN has produced several region proposals of varying sizes. Each proposal is represented using a five tuple (i,x,y,w,h), where i indicates the index of the proposal, and (x,y,w,h) are the coordinates of the rectangular detecting box (proposal). The ROI pooling layer has projected these proposals onto the feature matrix obtained from the last layer of Resnet50. The inputs to the ROI pooling layer are the feature matrix and the positive labelled proposals generated by the RPN. In the present work, '1000' ROI proposals out of total positive labelled proposals generated by the RPN have been selected by the ROI pooling layer. The value "1000" has been decided using *Bayesian optimization technique*. The ROI pooling layer has also performed the max-pooling operation on the feature matrix corresponding to each proposal generated by the RPN. Each proposal selected after ROI pooling layer has been fixed to 7 x 7 x 256 dimension and termed as ROI. There is one advantage of the proposed architecture is that all the ROI proposals generated from the RPN use same feature matrix obtained from the last layer of Resnet50.

4.4 Detection of textline

The proposed architecture in the present investigation has performed two different tasks—classification of the textline and a bounding box around the classified textline. The ROI obtained from the ROI pooling layer has been fed to the detection network (as shown in Fig. 10) one at a time. The detection network consists of two fully connected layers followed by the output layer. The output layer can act as a classifier to detect the textline as well as regressor to generate the rectangular box coordinates around the detected textline. The softmax function has been used as an activation function in the output layer to predict whether the textline is present or not. If the extracted probability score at the output layer of the detection network is greater than or equal to '0.5', then the output layer predicts the existence of the textline, otherwise the textline does not exist. If textline exists, then the regressor will give output of four coordinates of the rectangular box around this detected textline.

4.5 Training scheme

The proposed architecture has been trained using end-to-end method. During training, the document sample images from the training set have been provided as the inputs to the proposed architecture. The GT of each training sample has been developed manually. All the positive anchors generated from a single training image sample have been fed to the ROI pooling layer during training of the architecture. The loss is calculated as the sum of the difference between actual (i.e. GT) and predicted (i.e. output provided by detection network). If there is a textline present in the GT and the output layer classifier predicted it as "Not textline", then this will be counted as a loss.

In the proposed architecture, the total loss is the sum of the classification loss [26] and regression loss [26]. The classification loss measures the classification correctness of each predicted box. The regression loss measures the closeness of the coordinates of the box predicted by the proposed architecture around a classified textline. These two losses have been computed at the output layer of the detection network separately. The total loss calculated at the output layer of the detection network has been back propagated to the architecture for fine tuning. The classification loss and regression loss calculated during the training of the proposed architecture are presented in Fig. 11. The validation loss measures the total loss of the model on the validation set, whereas the training loss measures the total loss on the training set. The training loss and validation loss of the proposed architecture are compared with each other in Fig. 12.

5 Experimental results and analysis

The experimental evaluation of the proposed method has been carried out on the dataset mentioned in Section 3. The performance of the proposed method has been evaluated using the metrics *precision, recall* and *f-measure*. If the IoU value between the detecting and GT boxes is greater than 0.88 for any textline, then the detection has been considered as *true positive*, whereas *false positive* has been considered when more than one detecting boxes are generated on a single textline. *False negative* has been considered when no detecting box has been generated even when the textline is present in the document. The *precision*, *recall*, and *f-measure* graphs with respect to varying IoU values of the proposed TLS system are shown in Fig. 13. The *x*-axis denotes the IoU threshold and the *y*-axis denotes the performance of the proposed TLS system in terms of *precision*, *recall*, and *f-measure*.

Various batches of varying sizes have been formed of document image samples and the experiments have been carried out on these varying numbers of batch sizes. 'Epoch' denotes the total number of iterations performed to train the dataset. The experiments also have been carried out with varying numbers of epochs while keeping the batch size fixed. Table 4

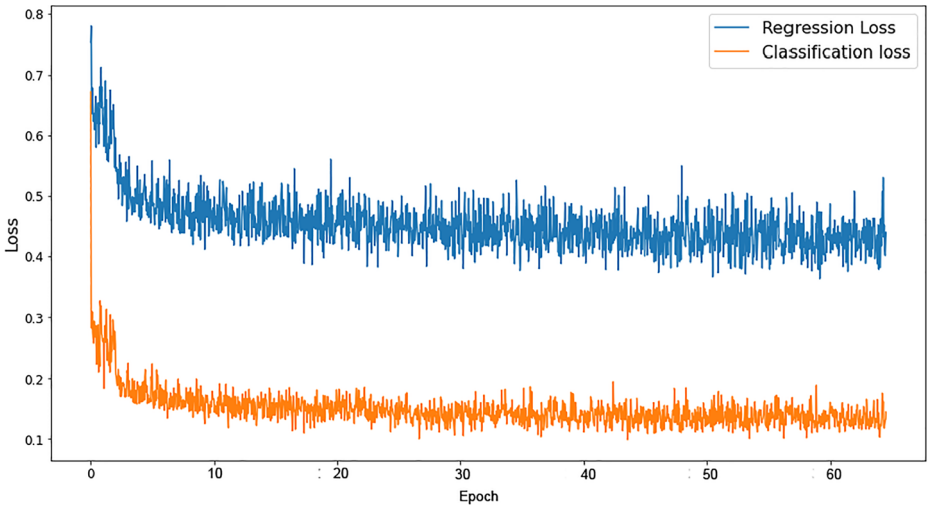


Fig. 11 Regression and classification loss during training

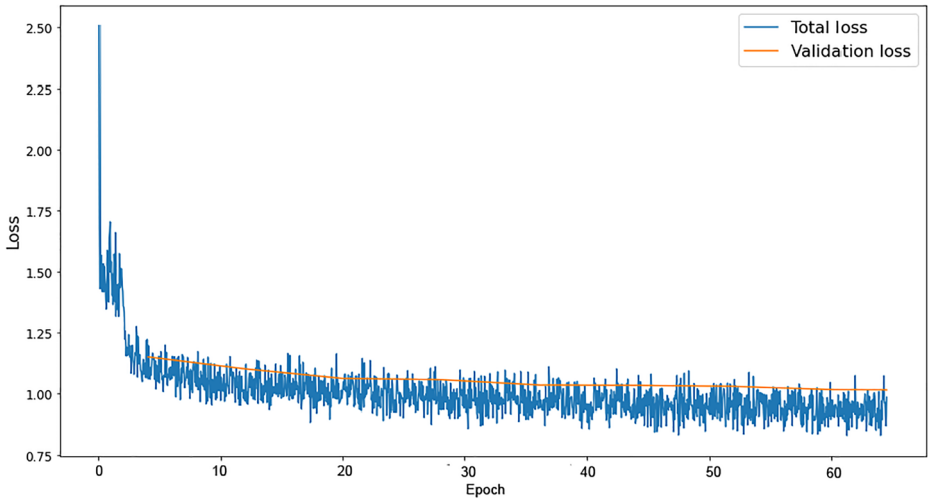


Fig. 12 Total loss compared with validation loss

shows the *precision*, *recall*, and *f-measure* values of the proposed TLS system with varying numbers of batch sizes and epochs.

The optimal set of values of various hyper parameters used in the proposed architecture of faster R-CNN is presented in Table 5. *Bayesian optimization technique* has been used to find this optimal set. A search space has been initialized for these hyper parameters. In the present investigation, the *f-measure* has been used as an objective function of the *bayesian optimization technique*. A sample vector has been selected from the search space to evaluate the cost of the objective function.

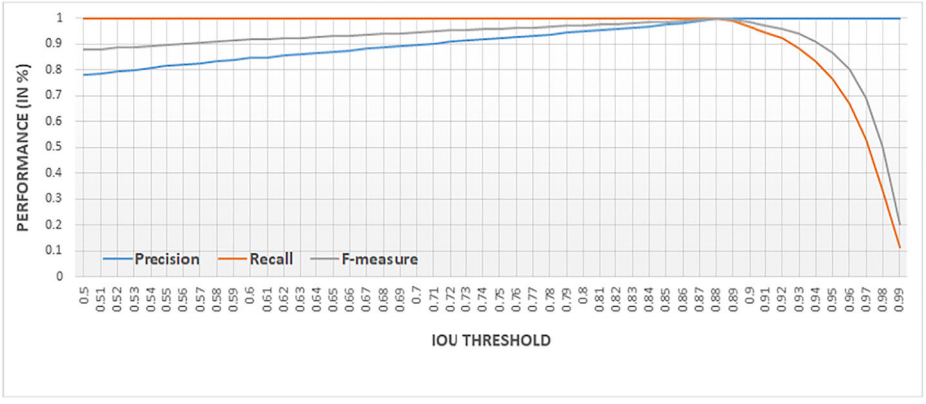


Fig. 13 Impact of varying IoU values on precision, recall and f-measure

Table 4 Precision, recall, and f-measure values of the proposed TLS system against various combinations of batch size and epoch

Batch size	Epoch	Precision	Recall	F-measure
2	25	100	96.15	98.04
2	50	100	99.96	99.98
2	75	100	99.72	99.86
4	25	100	98.92	99.46
4	50	100	96.09	98.01
4	75	100	95.54	97.72
8	25	100	95.67	97.79
8	50	100	94.93	97.40
8	75	100	94.93	97.40

5.1 Results with different backbone networks

The experiments have also been carried out by incorporating different backbone networks in the faster R-CNN architecture to have a performance comparison with the proposed architecture. Apart from Resnet50, five other backbone networks, namely, Resnet32, Resnet101, Resnet152, VGG-19 [16], and XceptionNet [2] have been incorporated in the faster R-CNN architecture to make a comparative performance evaluation. The TLS results in terms of *f-measure* with various backbone networks are shown in Table 6. In addition with these aforementioned backbone networks, the experiments have also been carried out on the dataset used in the present work using you only look oncev3 (YOLOv3) [25], spiking YOLO [12], and mask R-CNN [8] models of object detection. The spiking YOLO architecture has been generated by incorporating the spiking neural network model [32] in the YOLO object detection model. The TLS performances with YOLOv3, spiking-YOLO, and mask R-CNN are shown in Table 8.

The same optimal set of values of various hyper parameters mentioned in Table 5 has been used for the performance evaluation with other backbone networks as well. It has been observed that the network structures of Resnet101 and Resnet152 are very deep as compared to Resnet50. So, more document samples have been required to train these two networks, whereas Resnet32 is not able to capture the relevant features in comparison to Resnet50. The results from Table 6 demonstrate that the proposed faster R-CNN architecture with Resnet50 as a backbone network provides the best performance among all other backbone networks.

Table 5 Optimal set of values of various hyper parameter in faster R-CNN

Parameters	Search space	Step size	Optimal value
Learning rate	0.005 – 0.05	0.001	0.025
Gamma	0.03 – 0.05	0.01	0.05
Number of ROI proposals	500 – 5000	500	1000
IoU threshold	50% – 99%	1%	88%
Batch size	2 – 16	2	2
epoch	10 – 100	5	50

Table 6 Performance of the faster R-CNN with various backbone networks

Backbone networks	F-measure
Resnet32	78.25%
Resnet50	99.98%
Resnet101	91.20%
Resnet152	84.68%
VGG-19	79.68%
XceptionNet	88.23%

5.2 Qualitative result analysis

The performance of the proposed TLS method has been evaluated on 5953 test document samples of various manuscripts. Figures 14, 15 and 16 show the qualitative results obtained from the proposed method on three different test samples from ancient handwritten documents in Devanagari script.

Figure 14 shows the sample from Ayurveda manuscript containing one of the challenges dark spot present in the document, whereas Fig. 15 shows the sample from Tantra manuscript in which two textlines are written in skewed (another challenging factor) fashion. The Fig. 16 shows the qualitative result on an Ayurveda manuscript test sample containing another two challenging factors torn page and interline overlapping.

5.3 Training time

The experiments have been performed on 12GB NVIDIA Tesla K80 GPU with 2 cores of CPU system. The training times of the proposed system on varying numbers of batch sizes and epochs are presented in Table 7. If batch size increases, then the time to train the proposed architecture also increases.

5.4 Comparison with state-of-the-art results

Table 8 shows the performance comparison of the proposed TLS system with state-of-the-art results. The results in this table are presented in terms of *precision, recall, and f-measure*. As no dataset is publicly available on ancient historical handwritten document images in Devanagari script, so this work has developed a new dataset for the same. The performances

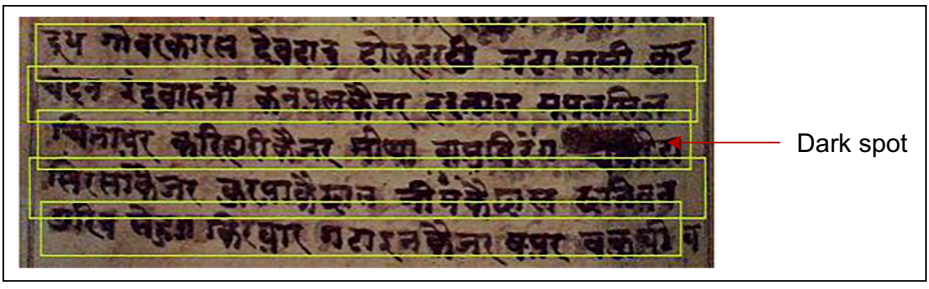


Fig. 14 Qualitative result on document having dark spot from Ayurveda manuscript. The green box indicates the detecting box generated by the proposed system

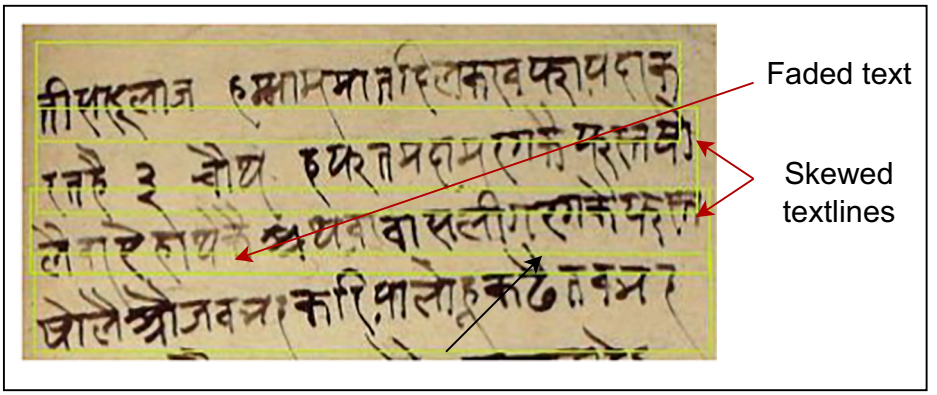


Fig. 15 Qualitative result on skewed text test sample from Tantra manuscript. This sample also contains the faded text. The green box indicates the detecting box generated by the proposed system

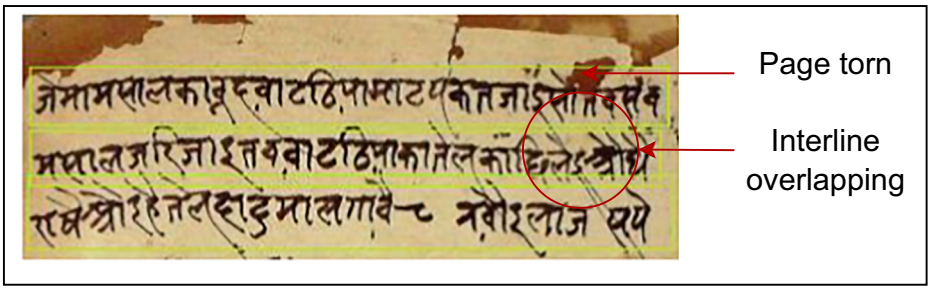


Fig. 16 Qualitative result on test sample having interline overlapped textlines and page torn from Ayurveda manuscript. The green box indicates the detecting box generated by the proposed system

Table 7 Training times of the proposed TLS system against various combinations of batch size and epoch

Batch size	Epoch	Training time (in minutes)
2	25	860
2	50	1810
2	75	2692
4	25	2230
4	50	3453
4	75	5340
8	25	2858
8	50	6660
8	75	8158

Table 8 Comparison with state-of-the-art results

Study	Script	Dataset	Method	Training samples	Performance
Narang et al. [21]	Devanagari	Used in the present work	Piecewise projection profile	13919	Precision- 93.26% Recall- 91.1% F-measure- 92.17%
Saabni et al. [29]	Devanagari	Used in the present work	Energy map seam carving	13919	Precision- 92.7% Recall- 96.01% F-measure- 94.33%
Garz et al. [3]	Devanagari	Used in the present work	Clustering grouping	13919	Precision- 96.57% Recall- 91.5% F-measure- 93.97%
Lenc et al. [14]	Devanagari	Used in the present work	ARU-Net	13919	Precision- 98.7% Recall- 95.35% F-measure- 97%
Renton et al. [27]	Devanagari	Used in the present work	FCN	13919	Precision- 92.43% Recall- 90.19% F-measure- 91.3%
Redmon et al. [25]	Devanagari	Used in the present work	YOLOv3	13919	Precision- 94.87% Recall- 93.31% F-measure- 96.43%%
Kim et al. [12]	Devanagari	Used in the present work	Spiking YOLO	13919	Precision- 97.34% Recall- 95.75% F-measure- 96.54%
He et al. [8]	Devanagari	Used in the present work	Mask R-CNN	13919	Precision- 97.87% Recall- 98.75% F-measure- 98.31%
Ren et al. [26]	Devanagari	Used in the present work	Conventional faster R-CNN	13919	Precision- 90.66% Recall- 68.62% F-measure- 78.12%
Proposed system	Devanagari	Self developed	Modified faster R-CNN	13919	Precision- 100% Recall- 99.96% F-measure- 99.98%

of the existing methods have been evaluated on the self-developed dataset used in the present investigation. It may be noted from Table 8 that, the proposed system has outperformed all the existing methods on TLS in ancient handwritten Devanagari text documents.

6 Conclusion and future work

The segmentation of the textlines in the ancient historical handwritten documents in Devanagari script is difficult due to the interline spacing, skewness, dark spots, faded text, etc. This investigation proposes a novel method of TLS in ancient handwritten document images in Devanagari script. The proposed system overcomes various drawbacks found in existing TLS systems based on projection profile and grouping methodology. The proposed system is capable to detect the textline correctly even if the documents are faded, or the dark spots present on the textlines or the interline overlapping and skewness exist in the documents. The proposed system is more cost effective because it uses the RPN algorithm to generate the region proposals instead of selective search method. The results demonstrate that the proposed system outperforms the state-of-the-art TLS methods as well as the conventional faster R-CNN based method of TLS in ancient historical handwritten Devanagari document images. The major drawback of the proposed system is when it detects any skewed textlines for few samples the detecting box of the textline overlaps some text portions of the previous and next textlines. Another limitation of the proposed system is it cannot detect the textlines correctly for few samples if interline overlapping exist.

This work can be extended to segment the textlines in various ancient historical documents in other Indian as well as non-Indian scripts. The attempt will also be made in future to perform the word and character level segmentation in the ancient handwritten document images. Among other possible directions, it has been planned to extend this work up to the recognition of each handwritten word in the entire ancient document in Devanagari script. Another possible future direction of this work is to recognize the handwritten text in the ancient historical documents in other non-Indian scripts as well.

Declarations

Conflict of Interests The authors have no conflict of interest/competing interest to declare that are relevant to the content of this article.

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